

have the same execution strategy as trading two 500,000 share orders with two different VWAP algorithms (provided that the algorithms are equivalent). Additionally, trading 1,000,000 shares with one broker's POV algorithm (e.g., $POV = 20\%$) will have the same execution strategy as using two different broker POV algorithms at one-half the execution rate (e.g., $POV = 10\%$ each). A paired observation approach ensures that identical orders are executed under identical market conditions. Analysts can also choose between the arrival cost and VWAP benchmark as the performance metric. Our preference for the paired sample tests is to use the VWAP.

An independent sampling approach is used to compare orders that are executed over different periods of time using different algorithms. This test is appropriate for algorithms such as implementation shortfall that manage the trade-off between cost and risk and employ dynamic adaptation tactics. In these cases we do not want to split an order and trade in algorithms that adapt trading to real-time market conditions because we do not want these algorithms to compete with one another. For example, if a 1,000,000 shares order is split into two orders of 500,000 shares and given to two different brokers, these algorithms will compute expected impact cost based on their 500,000 shares not on the aggregate imbalance of 1,000,000 shares. This is likely to lead to less than favorable prices and higher than expected costs since the algorithms will likely transact at an inappropriately faster or slower rate. The algorithm may confuse the incremental market impact from the sister order with short-term price trend or increased volatility, and react in a manner inappropriate for the fund, resulting in higher prices. Our preference is to use the arrival cost as our performance metric in the independent sample tests.

A paired observation approach can use any of the static algorithms providing that the underlying trade schedule is the same across brokers and algorithms, e.g., VWAP and POV. An independent sampling approach needs to be used when we are evaluating performance of dynamic algorithms that adapt to changing market conditions.

Non-Parametric Tests

We provide the outline of six non-parametric tests that can be used to determine if two algorithms are equivalent. They are based on paired samples (Sign Test, Wilcoxon Signed Rank Test), independent samples (Median Test, Mann-Whitney U Test) and evaluation of the underlying data distributions (Chi-Square and Kolmogorov-Smirnov goodness of fit). Readers who are interested in a more thorough description of these tests as